SIGMIDI FOOT CONTROLLER

v0.1 / UNIFIED ELECTRICAL SCHEMATIC / external board pins only

Continuous wires connect every fitted terminal. Dots = junctions; arched crossings = no connection. Component placement is schematic.

Nano and LCD pins keep physical order. Nano: JP1 left 1-15, JP2 right 15-1, USB-C up. Outlines are schematic.

3V3 / Nano logic supply

+5V LED

J1 5.5 x 2.1mm
9V DC / center-negative
SLEEVE (+)
CENTER (-)

C1
47uF / 35V

U3 S13V10F5
Pololu #4083 / 1178565
VIN
GND
5V OUT

J2 ACJS-MHDC / passive pedals
TOP / pedal 1
11 RING
12 TIP
10 SLEEVE
BOTTOM / pedal 2
5 RING
6 TIP
4 SLEEVE
1/2/3/7/8/9: NC; pad 13: mechanical / NC

R20 5.1k
R30 5.1k

USB-C / MIDI

U1 Arduino Nano ESP32
Component-side top view / USB-C up

JP1:1	D13	JP2:15 / GPIO47	D12
JP1:2 / 3V3	3V3	JP2:14 / GPIO38	D11
JP1:3	B0	JP2:13 / GPIO21	D10
JP1:4 / GPIO1	A0	JP2:12 / GPIO18	D9
JP1:5 / GPIO2	A1	JP2:11 / GPIO17	D8
JP1:6	A2	JP2:10 / GPIO10	D7
JP1:7 / GPIO4	A3	JP2:9 / GPIO9	D6
JP1:8 / GPIO11	A4	JP2:8 / GPIO8	D5
JP1:9	A5	JP2:7 / GPIO7	D4
JP1:10	A6	JP2:6 / GPIO6	D3
JP1:11	A7	JP2:5 / GPIO5	D2
JP1:12 / VBUS	VBUS / VUSB	JP2:4 / GND	GND
JP1:13	B1	JP2:3	RST
JP1:14 / GND	GND	JP2:2	D0 / RX0
JP1:15 / VIN	VIN	JP2:1	D1 / TX0

All printed labels retained; unused pads left unwired.

U2 Waveshare 24382
1 BL
2 RST
3 D/C
4 CS
5 SCL / CLK
6 SDA / DIN
7 GND
8 VCC
P2 / GH1.25 / SPI / pins 1-8

SW11 ENCODER
A
20 detents + click
B
Internal pull-ups
SW1
No external R/C
C
SW2

USB VBUS sense / 3.0V at 5V
R50 100k
R51 150k

R21 10k
R31 10k

Pololu 2535 / optional LEDs

LED1 / B	2535	DIN, 5V, GND, DOUT
LED2 / A	2535	DIN, 5V, GND, DOUT
LED3 / 2	2535	DIN, 5V, GND, DOUT
LED4 / 1	2535	DIN, 5V, GND, DOUT
LED5 / C	2535	DIN, 5V, GND, DOUT
LED6 / 3	2535	DIN, 5V, GND, DOUT
LED7 / D	2535	DIN, 5V, GND, DOUT
LED8 / 4	2535	DIN, 5V, GND, DOUT
LED9 / E	2535	DIN, 5V, GND, DOUT
LED10 / 5	2535	DIN, 5V, GND, DOUT (NC)

MODULAR FOOTSWITCH LADDER

R7 10k	R9 1k	R10 1k	R11 1k	R12 1k	R13 1k	R14 1k	R15 1k	R16 1k	R17 1k	R18 1k
R8 20k	C5 100nF	SW1 B	SW2 A	SW3 2	SW4 1	SW5 C	SW6 3	SW7 D	SW8 4	SW9 E	SW10 5
		T1 / LED1	T2 / LED2	T3 / LED3	T4 / LED4	T5 / LED5	T6 / LED6	T7 / LED7	T8 / LED8	T9 / LED9	T10 / LED10

COMMON GND

PASSIVE PEDALS

Pot: outer legs RING / SLEEVE; wiper TIP.
Variable resistor: TIP to SLEEVE; RING unused.
0-10k per input. Bias HIGH in resistor mode;
high impedance in pot / disabled / boot states.

ASSEMBLY

Set 4 / 6 / 8 / 10 switches; calibrate installed taps.
SW11 contact pin arrangement is functional.
Native USB-C carries USB MIDI and ROM flashing.
USB-only supply does not power the external LED rail.

LED PACKAGE

Lens left, leads right, flat side down.
Top to bottom: DIN / 5V / GND / DOUT.
Only LEDs and wires; direct D7 / GPIO10 data.
Check 3.3V data acceptance on the actual LED batch.

Netlist, BOM and pin references: design/electrical/. All 46 components and 41 nets checked against the connected drawing.